

A 112-bit BeiDou Rescue-Telemetry Node: From Survivor Position to UAV Mission

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BDS-SMC2 · Dissertation Objective 5 — RTK-corrected survivor location injected into BeiDou short messaging (Class + PNT)

Node: **BDS-SMC2**

ESP32 + BDS-3 RDSS (EVBKIT_V3)

Portal: bdrd.hwasmart.com

github.com/letsoalomaile1/BDS-SMC2

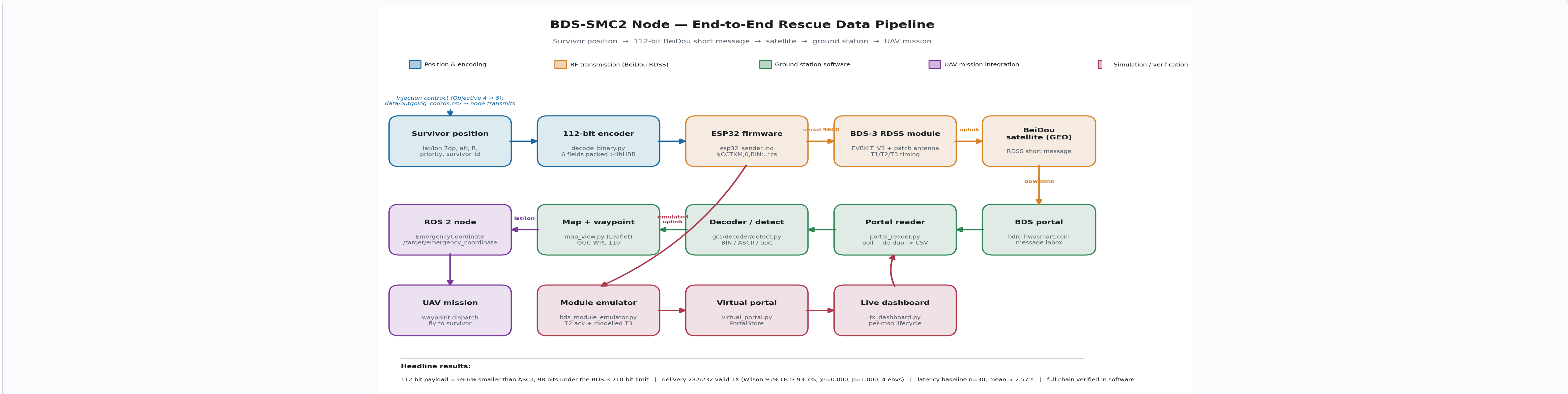


Figure 1 — End-to-end node pipeline. Every block is implemented: survivor position → 112-bit encoder → ESP32 firmware → BDS-3 RDSS module → BeiDou satellite → portal → reader → decoder → map / waypoint → ROS 2 EmergencyCoordinate → UAV mission. The lower (red) branch is the simulation harness that verified the full chain while the RF uplink was blocked.

1 · Problem & Motivation

In disaster zones terrestrial networks (GSM/LoRa) fail and satellite messengers (Iridium, COSPAS-SARSAT) carry no custom payload. **BeiDou Short Message Communication (BDS-SMC / RDSS)** needs no ground infrastructure and supports a 210-bit user payload — but its use as a structured *rescue-telemetry* link for UAVs is uncharacterised.

This node answers one question: **can a complete six-field rescue payload fit a single BDS-3 short message and travel reliably, end-to-end, into a UAV mission pipeline?**

2 · Objectives & Research Gaps

Gap	Question	Status
1	ASCII vs binary coordinate encoding	Complete
2	End-to-end latency (T1/T2/T3)	Baseline
3	Delivery rate across environments	Complete
6	Telemetry compression (ASCII/Huffman/binary)	Complete

Objective 5: emulate a survivor's RTK-corrected position and inject it into BDS short messaging as Class + PNT data, then demonstrate it driving a UAV waypoint.

3 · Methods — Hardware & Encoding

- Platform:** ESP32-D0WD-V3 dev board driving a BDS-3 RDSS module (EVBKIT_V3) with on-board patch antenna, via an RS232–TTL adapter.
- Frame:** \$CCTXM,<destID>,<content>*cs (NMEA-style XOR checksum).
- 112-bit payload** packs six fields (>iihHBB): lat / lon (int32, 7 dp ≈ 1 cm), altitude (int16 m), uncertainty R (uint16 cm), priority, survivor ID.
- Timing:** firmware tags T1 (send), T2 (module ack), T3 (delivery) for per-message latency.

4 · Methods — Measurement & GCS

- Field protocol:** 4 environments (open sky, light canopy, urban canyon, indoor) × 3 locations × repeated TX; serial logger writes every frame to CSV.
- Analysis (Python):** chi-square test of delivery homogeneity and Wilson 95% confidence intervals (SciPy); bit-accounting for encoding gaps.
- Ground station (GCS):** portal reader → decoder (BIN/ASCII/text) → Leaflet map + QGC WPL 110 waypoint → ROS 2 EmergencyCoordinate on /target/emergency_coordinate.
- Simulation harness:** a virtual BeiDou link (module emulator → virtual portal → live dashboard) reproduces byte-identical frames and Gap-2 latency, verifying the chain without RF.

5 · Results — Key Findings

−57.6% rescue payload vs ASCII (264 → 112 bits)	98 bits headroom under the 210-bit BDS-3 limit
232/232 valid TX received (Wilson 95% LB ≥ 93.7%)	≈2.57 s mean latency, ASCII baseline (n = 30)

Encoding (Figs 2–3): binary packing is 57.6% smaller than ASCII for the rescue payload and 69.6% smaller than a full ASCII telemetry string, beating Huffman while carrying more fields. **Delivery** (Figs 4–5): 100% across all four environments and all 12 sites; $\chi^2=0.000$, $p=1.000$. **Latency** (Fig 6): every baseline message lands inside the 5 s budget (median ≈ 2.5 s); full diurnal sessions on the 112-bit payload remain to be collected.

6 · Discussion & Limitations

- Full pipeline is **software-complete and verified end-to-end** via the simulation harness; frames are byte-identical to firmware output.
- Position is **emulated** (lab ground-truth T001–T006), as Objective 5 specifies; live RTK is a separate work item.
- RF uplink hardware bring-up is **unresolved** — one TX must confirm the module accepts the 28-char hex payload.
- Link is **one-way**; single hardware unit (no generalisability claim); 100% delivery reported as Wilson lower bound.

7 · Conclusion & Future Work

A complete six-field rescue payload fits a single 112-bit BDS-3 short message with **98 bits to spare**, delivers at ≥ **93.7%** (Wilson LB) with ~**2.6 s** latency, and drives a UAV mission through a working GCS pipeline.

Next: hardware TX verification · full diurnal latency on the 112-bit payload · live RTK position · group EmergencyCoordinate.msg field extension.

References

- Li G. et al. (2021). Introduction to the global short message communication service of BeiDou-3. *Advances in Space Research* 67(5), 1701–1708. doi:10.1016/j.asr.2020.12.011.
- BeiDou RDSS user portal — bdrd.hwasmart.com.

Results — Figures

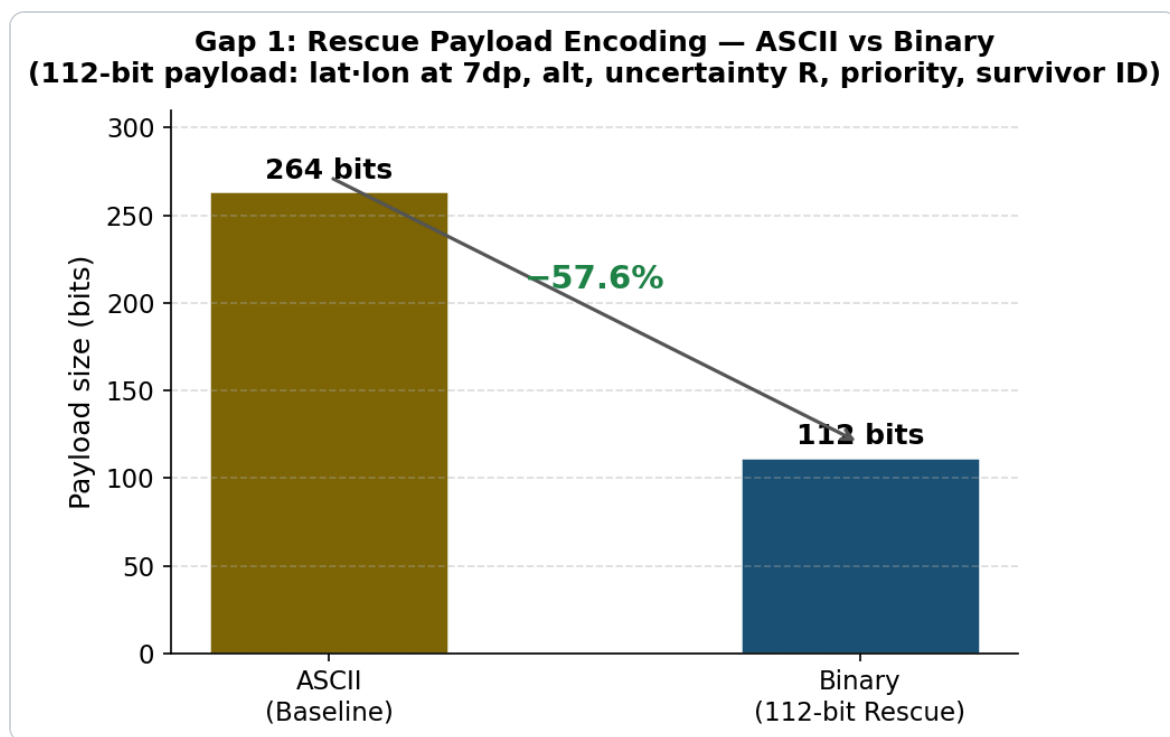


Fig 2 — Gap 1: rescue payload 264 → 112 bits (−57.6%).

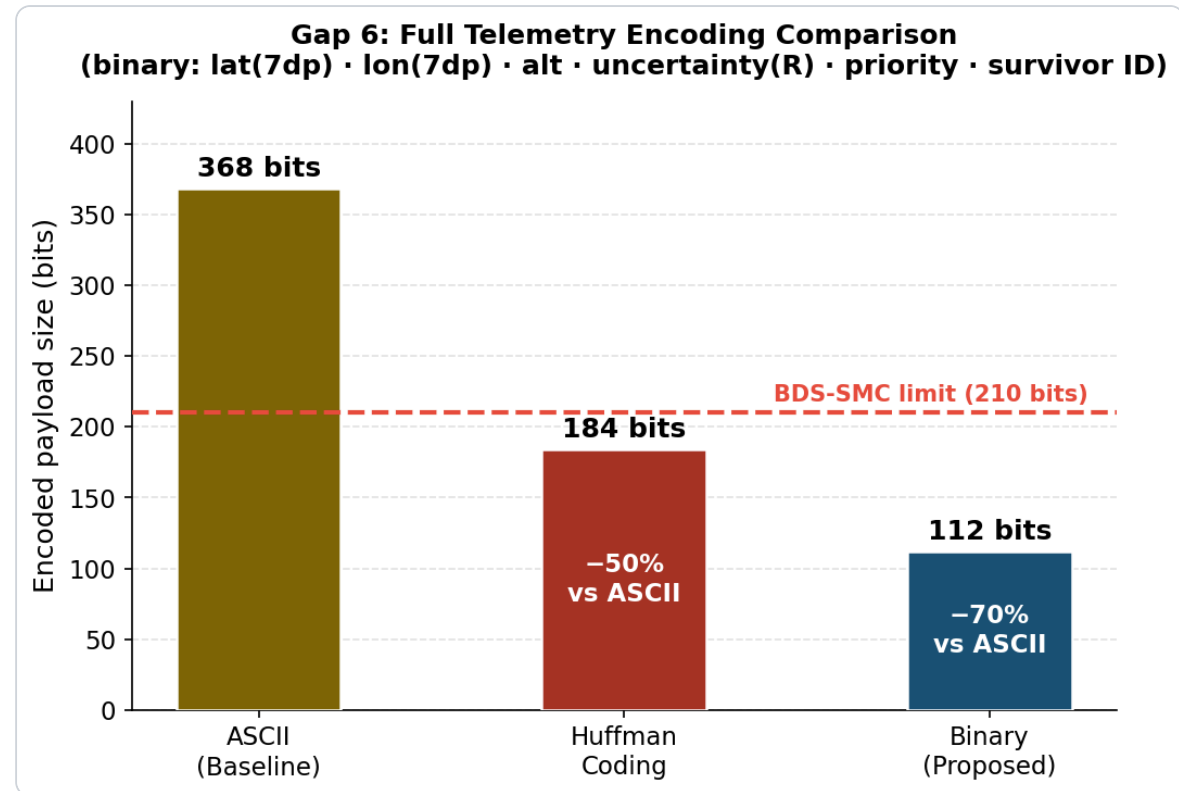


Fig 3 — Gap 6: binary vs ASCII vs Huffman (−69.6%).

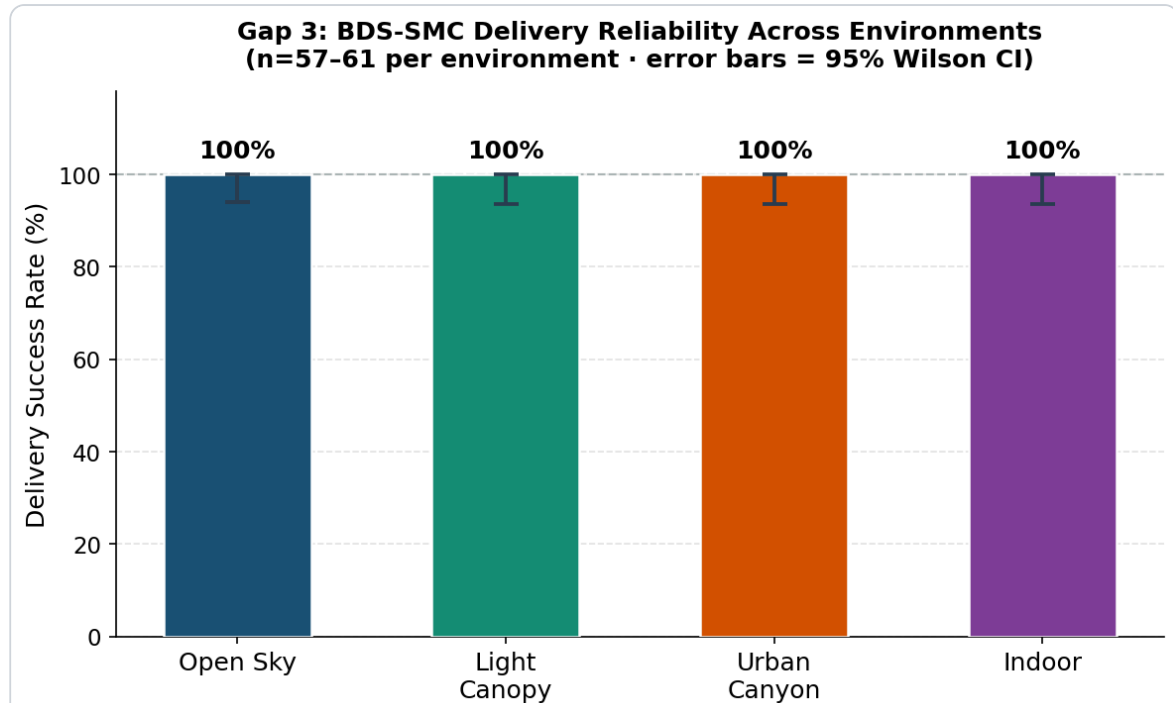


Fig 4 — Gap 3: 100% delivery, 4 environments.

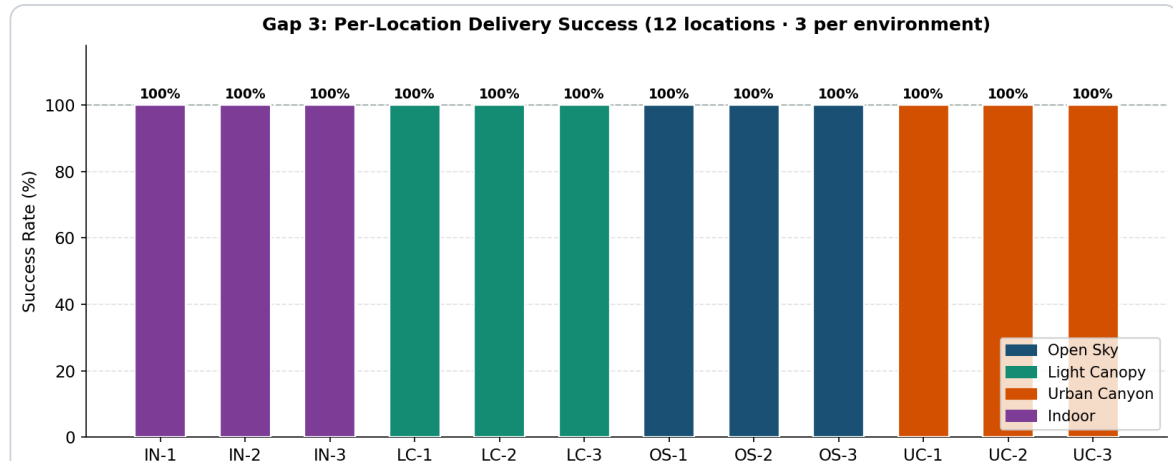


Fig 5 — Gap 3: all 12 sites at 100%.

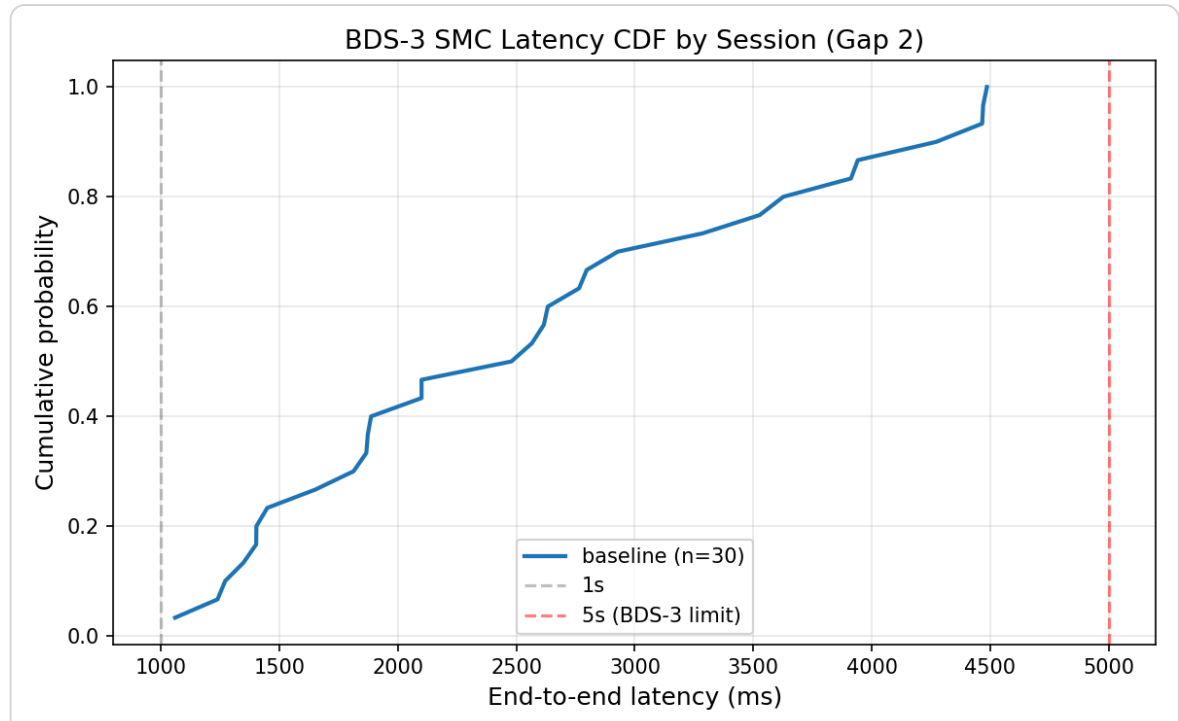


Fig 6 — Gap 2: latency CDF inside 5 s budget.

8 · Live Demonstration — GCS Dashboard

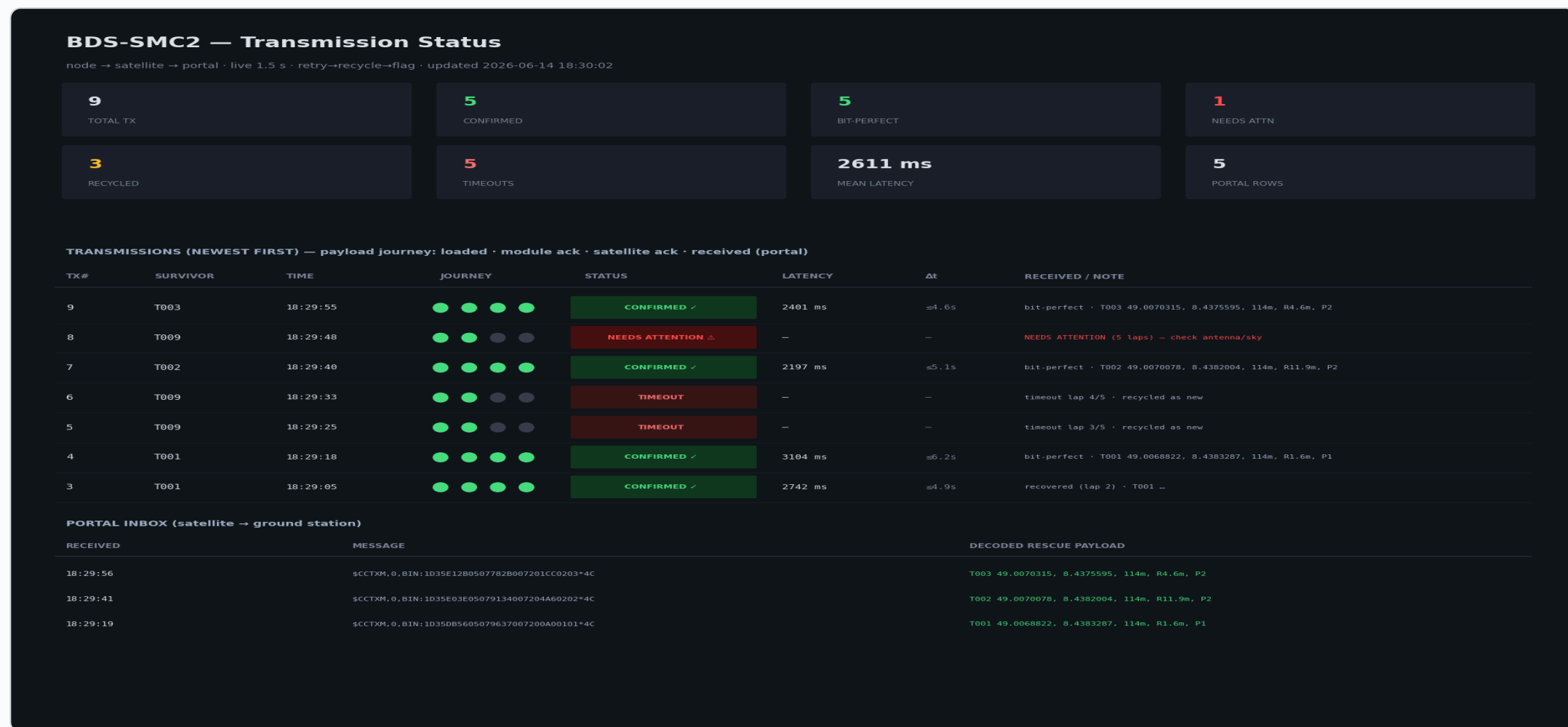


Figure 7 — Real-time transmission dashboard: per-message lifecycle (loaded → module ack → satellite ack → received at portal), live latency, and the decoded rescue payload for each survivor. Verified end-to-end against the simulation harness with byte-identical frames.